

## Problem 1

1a

The system is Controllable and Observable.

1b

The resulting optimal state-feedback gain is  $K = [2.7016 \quad 3.3227]$ . The closed-loop poles of the system matrix  $A - BK$  are  $\lambda = \{-1.0506, -6.0948\}$ .

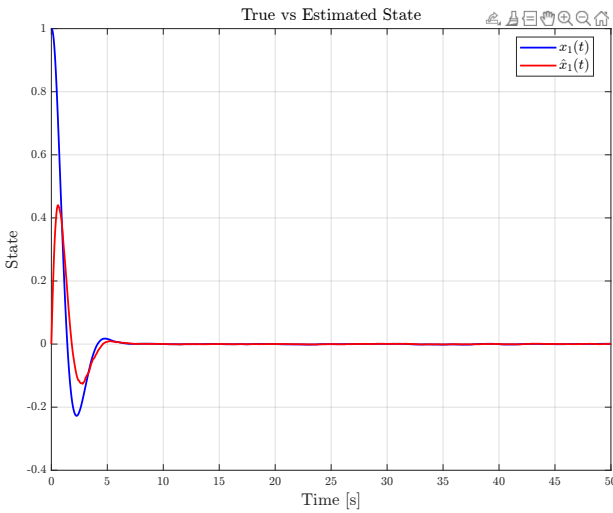
1c

The optimal steady-state Kalman gain is  $L = \begin{bmatrix} 1.4066 \\ 0.9893 \end{bmatrix}$ .

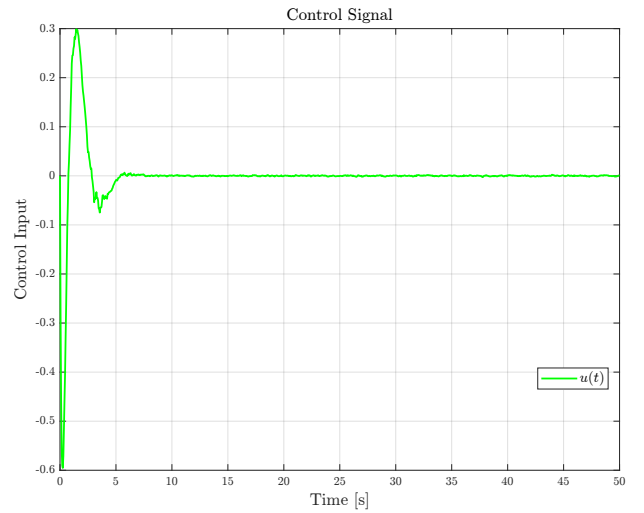
1d

The closed-loop system matrix combining LQR and Kalman filter is  $A_{\text{LQG}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & -0.5 & -5.4032 & -6.6454 \\ 1.4066 & 0 & -1.4066 & 1 \\ 0.9893 & 0 & -7.3925 & -7.1454 \end{bmatrix}$ .

1e



(a) Comparison between the true state  $x_1(t)$  and its estimate  $\hat{x}_1(t)$  produced by the steady-state Kalman filter.



(b) Optimal control input  $u(t)$  applied by the LQR regulator based on the estimated state.

Figure 1: Simulation results of the closed-loop LQG regulator system over 50 seconds. (a) shows how the observer tracks the true state under noisy conditions. (b) illustrates the corresponding control effort computed using the LQR law based on the observer estimate.

## Problem 2

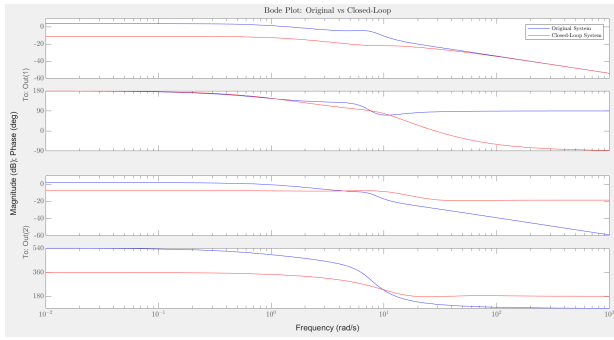
2a

The  $\mathcal{H}_\infty$  norm from disturbance  $v_1$  to outputs  $y_1, y_2$  is  $\|G\|_\infty = 4.8509$ . This peak gain occurs at a frequency of  $\omega = 6.7473$  rad/s.

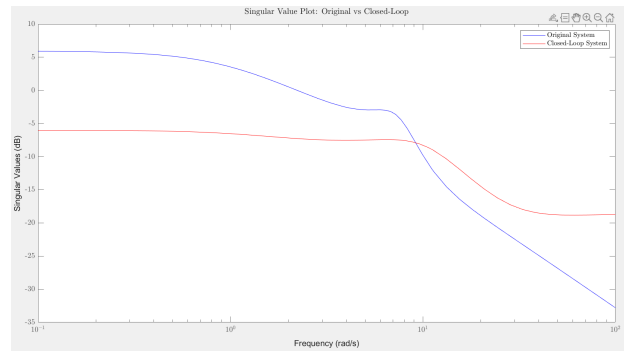
2b

The achieved  $\gamma$  value is 0.4063, which equals the  $\mathcal{H}_\infty$  norm of the closed-loop system. The peak occurs at  $\omega = 1.8226$  rad/s.

2c



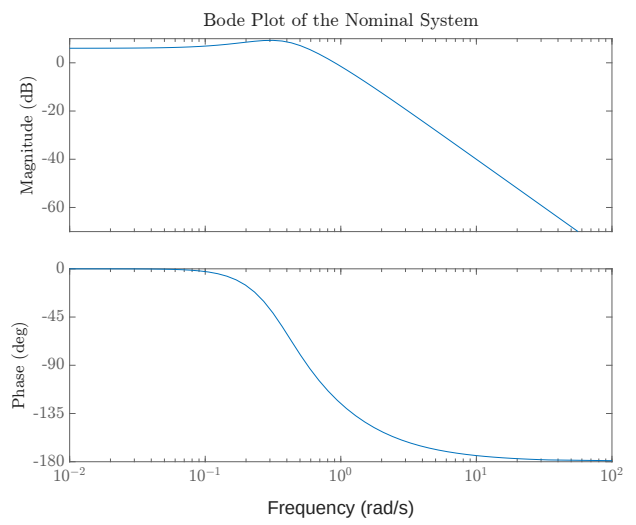
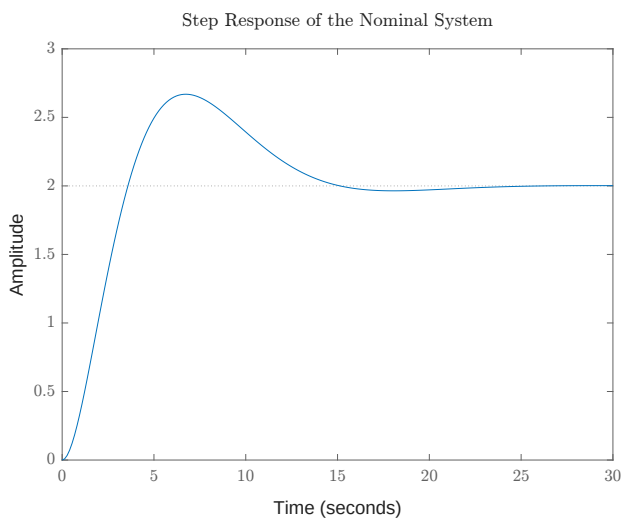
(a) Bode plot



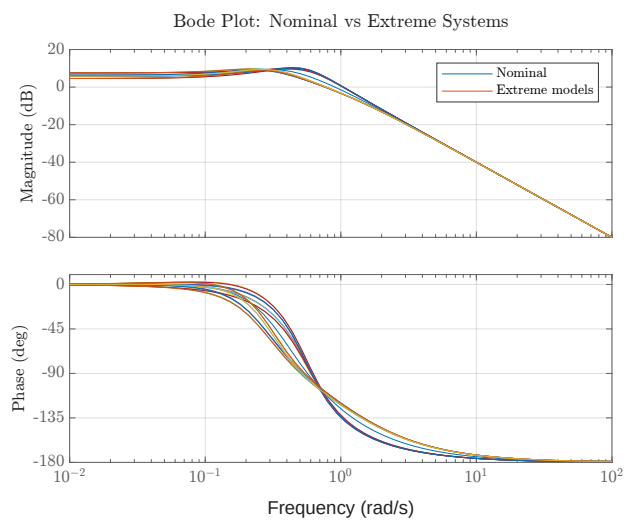
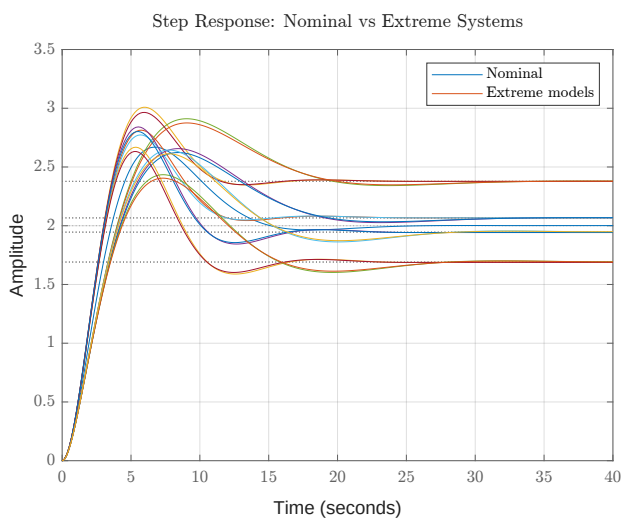
(b) Singular value plot

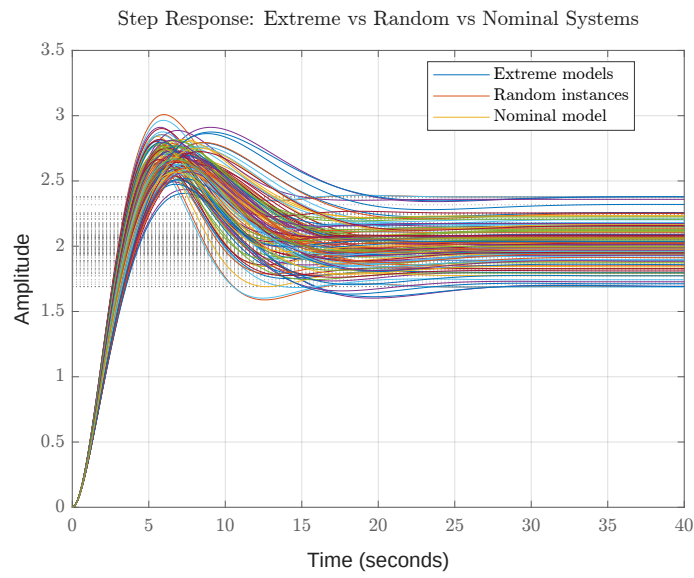
## Problem 3

3a



3b





The limits of the system's dynamic performance are well captured by the extreme models in Part (b). These models illustrate the worst-case situations by testing the system with both the higher and lower extremes of each unknown parameter. The random examples in Part (c), on the other hand, reflect a greater range of behavior, demonstrating that the extreme models, and is useful for capturing limitations, but might not adequately capture the range of potential system responses.

**Thank You**